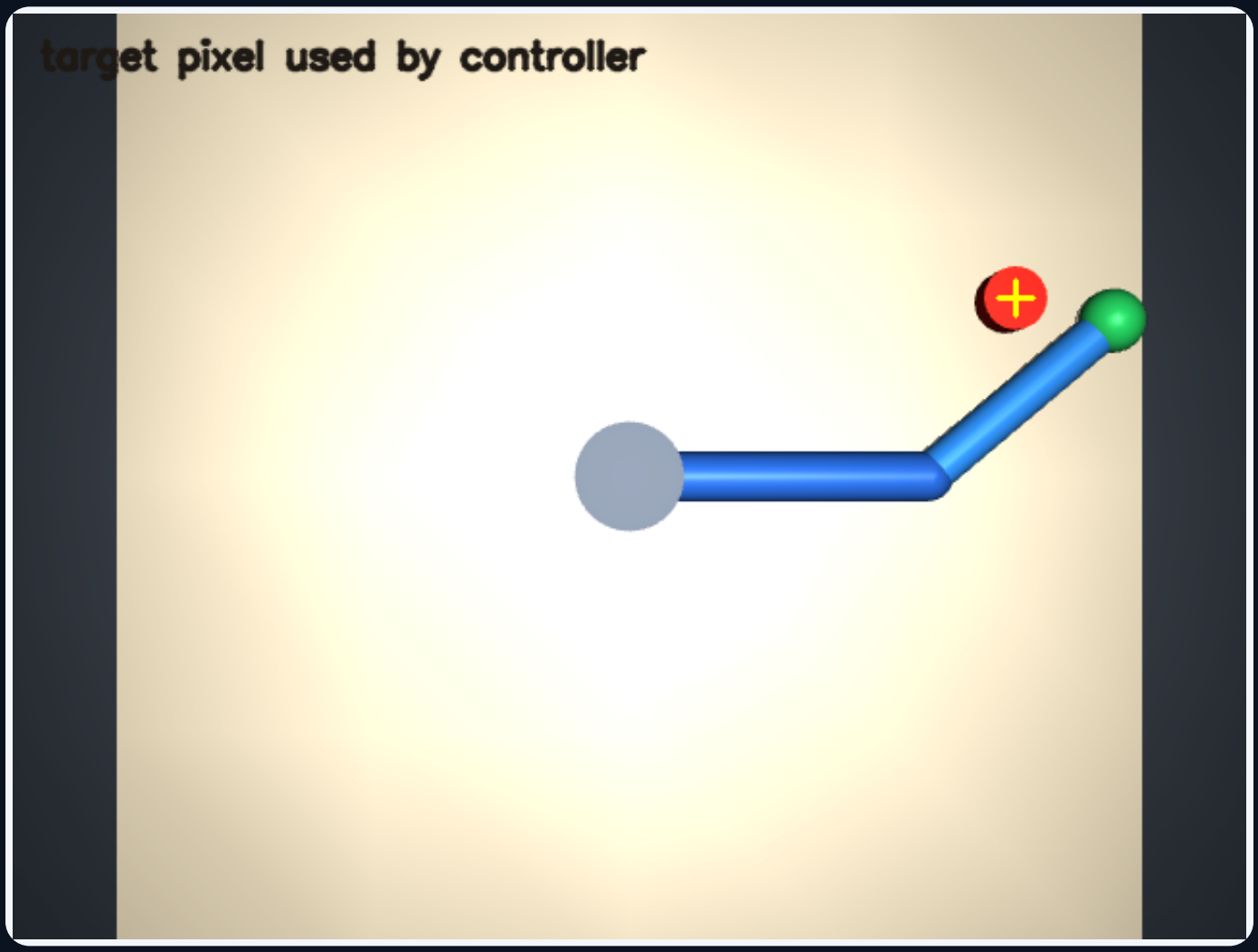
ROBOTICS PORTFOLIO | PROJECT GUIDE

Camera-Based Visual Servoing

A project explainer for visual-servo-mujoco



Pixels -> a table-plane estimate -> joint motion -> a measured result

This guide explains the important code, what the benchmarks actually establish, which experiments were rejected, and the next evidence-based milestone.

MuJoCo | OpenCV | Image Jacobian | Paired benchmarks

Local work on feat/robustness-benchmark | Updated 16 Oct 2026

The system in one picture

A small simulation that makes every perception and control step inspectable.

01

Render

MuJoCo produces a top-down RGB image of the table, target, and arm.

02

Detect

OpenCV segments the red target and green end-effector marker.

03

Estimate

A pinhole camera model maps target pixels to table coordinates.

04

Move

Inverse kinematics or image feedback updates the two arm joints.

05

Score

MuJoCo ground truth measures final distance after control stops.

What is measured and what is not

The controller receives rendered pixels and joint state. Ground-truth target position is read after the run for scoring; it is not fed into the controller.

What one feedback update does

Image feedback corrects pixel error with a local model of how each joint moves the image.

DAMPED IMAGE-SPACE UPDATE

$$e = target_pixel - tip_pixel$$

$$dq = 0.5\ J^T (J\ J^T + I)^{-1}\ e$$

then clip each joint change to 0.30 rad and advance 0.4 simulated seconds

1. Measure

Find red target and green tip pixels.

2. Compare

Compute the 2D image-space error.

3. Solve

Use the Jacobian and damping term to get dq.

4. Advance

Move, render again, and count a fresh check.

Why the damping term is there

The arm has two joints but the observed error has two pixel coordinates. Near a singular pose, a raw inverse can demand a huge joint change. Adding I stabilizes the solve, and the per-joint cap limits one update.

Two different tip sources in experiments

The standard image-feedback controller detects the tip in the camera image. The timing benchmark can instead project the current joint angles through the calibrated geometry. That is current proprioceptive state paired with a delayed target image, not fully delayed vision.

Important constants in run.py

8 px tolerance | 3 consecutive checks | 20 maximum motion updates | 5 missing-target observations of grace | 4.5 cm physical success radius

The controller stop reason and the physical reach score are separate fields in each trial.

Where the important code lives

The core loop is intentionally small. Benchmarks isolate one assumption at a time.

model.py — MODEL_XML

Inline scene: camera, table, two-link arm, target, and position actuators.

controller.py — projection + IK + Jacobian

Camera geometry, forward/inverse kinematics, pixel Jacobian, damped joint update, and opt-in predictor functions.

run.py — run_trials

Detection, seeded targets, control loop, dropouts, camera delay, visibility trace, stop reason, physical scoring, and demo recording.

benchmark.py — paired robustness suite

Same seeded targets across open loop and image feedback; aggregates success and error percentiles.

dynamic_benchmark.py — moving target + latency

Separates target motion, observation age, and strict vs experimental stopping.

prediction_benchmark.py / reversal_benchmark.py — opt-in predictor tests

Compares measured target with extrapolators while keeping current joint-state tip projection explicit.

sensor_timing_benchmark.py — feedback source test

Compares fresh RGB, fully delayed RGB, and delayed target plus current-state tip.

verification_*.py / terminal_verification.py — fresh-view evidence + quality

Renders a fresh terminal view, scores marker quality, and measures calibration sensitivity, coverage, and decision errors.

camera_calibration_*.py — landmark health + transfer

Matches fixed landmarks to pinhole projections, freezes a threshold, then evaluates pose transfer, visibility, and radial distortion.

Good first files to read

Start with model.py, then controller.py, then run.py around run_trials and the image_feedback branch. Read benchmark modules after you can trace one episode.

What the current controller handles

Image feedback corrects several errors, but noisy sensing and visibility still expose real failure modes.

PHYSICAL SUCCESS BY TARGET-PIXEL NOISE



Blue: open loop Green: image feedback | 100 trials per condition

At 20 px noise

The default feedback controller reached 88/100 targets within 4.5 cm; open loop reached 34/100. Feedback median error was 2.12 cm and P95 was 6.39 cm. The controller uses a three-sample median, but this does not eliminate large-noise failures.

DROPOUT LENGTH SEPARATES RECOVERY FROM TIMEOUT

Injected blackout	Physical success	Reacquired after gap
1 observation	100/100	100/100
3 observations	100/100	100/100
5 observations	100/100	100/100
6 observations	33/100	0/100

Ground-truth distance decides physical success. The controller can stop without fresh vision even when the arm is physically close.

Natural self-occlusion is the next measured gap

In the clean strict run, 91/100 trials reached physically but stopped after the arm hid the target from the overhead camera.

Three sensor setups that must stay distinct

The timing experiments separate image age from the source used to locate the arm tip.

Fresh RGB

target pixel now | tip pixel now

Fully delayed RGB

target pixel delayed | tip pixel delayed

Vision + proprioception

target pixel delayed | tip from current joints

How to read the held-out results

Each row uses seeds 100-104 and 100 trials. The delta is paired median reaching error: hybrid minus fully delayed RGB. A negative value means lower error in this simulator.

Target path	Noise	Delay	Delayed RGB	Hybrid tip
Sinusoidal	0 px	0.4 s	4%	84%
Sinusoidal	0 px	0.8 s	63%	81%
Sinusoidal	2 px	0.4 s	11%	68%
Sinusoidal	2 px	0.8 s	56%	85%
Piecewise linear	0 px	0.4 s	34%	88%
Piecewise linear	0 px	0.8 s	58%	93%
Piecewise linear	2 px	0.4 s	37%	83%
Piecewise linear	2 px	0.8 s	56%	95%

What improved, and what did not

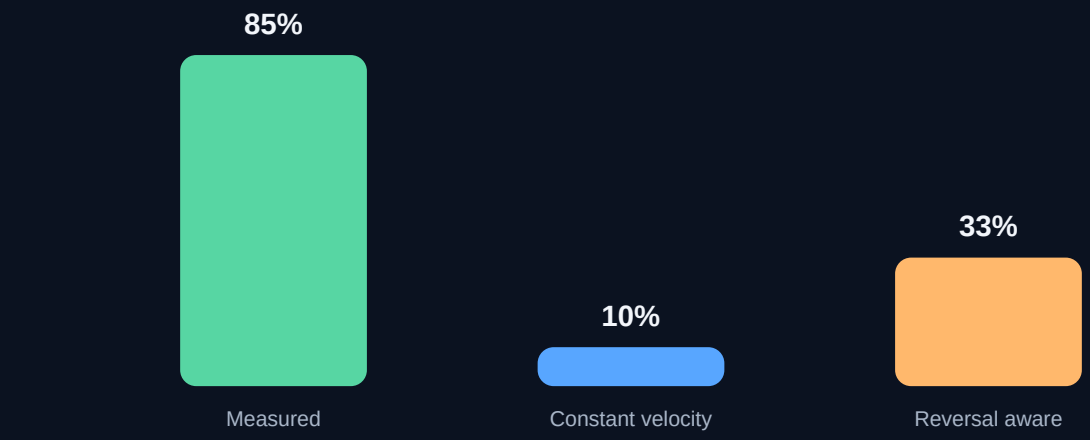
The current-joint-state tip hybrid improved physical success in all eight held-out settings. Its P95 error was 4.92-6.71 cm. It still had 0/100 current-time visual confirmations in every delayed condition. Physical reaching and fresh visual completion remain separate outcomes.

The paired bootstrap resamples only five seed groups. The interval reflects these simulator runs, not broad generalization.

A useful negative result: prediction lost to measurement

Moving-target latency made extrapolation tempting. Paired evidence says these predictors stay experimental.

SINUSOIDAL TARGET | 0.4 S DELAY | NOISE-FREE



Success within 4.5 cm | five seeds x 20 targets

Method	Median reaching error	Success
Measured target	2.70 cm	85%
Constant velocity	6.48 cm	10%
Reversal aware	6.44 cm	33%

Across all eight reversal benchmark settings

The reversal-aware predictor increased median error in every setting; each seed-cluster bootstrap interval for the paired error delta was above zero. It often beat plain velocity prediction, but never beat the measured-target baseline on success. Keep predictors opt-in and preserve the baseline.

At 0.8 s delay the reversal-aware method reached only 5% on sinusoidal noise-free trials versus 85% for measured target. The measured result rules out this simple reversal guard as the default latency fix.

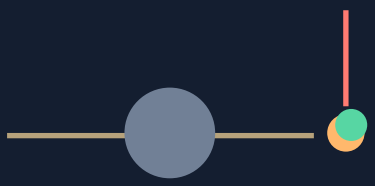
Other measured tradeoffs

The three-sample median helped 20 px noise versus raw feedback, while larger windows raised clean error. Stale-target stopping did not improve physical success and can stop without reacquisition. These remain scoped experiments, not new defaults.

A second view can supply a separate visual estimate

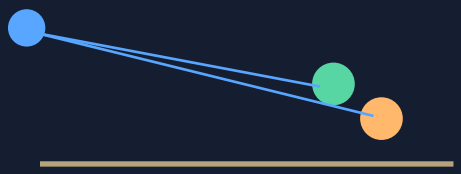
Four oblique candidates were evaluated; an opt-in east-view check now runs after the overhead controller stops.

OVERHEAD CONTROL VIEW



The arm can cover the target from above.

EAST OBLIQUE CANDIDATE



Pixels are projected to known marker heights.

View	Control noise	Both seen	False success	Missed success
East	0 px	200/200	0	1
East	20 px	198/200	0	14
North	0 px	200/200	0	0
North	20 px	200/200	6	11

Why east is the current candidate

Across the two seed groups it produced no false success estimates and saw both markers in 198/200 noisy trials. It missed 14 true noisy-condition successes. The north view saw every marker but made six false success estimates. The conservative tradeoff is promising, not sufficient for a production stop rule.

Scope of this evidence

The oblique camera is rendered from the simulator. The controller still uses the overhead camera, and terminal verification remains a separate fresh observation after control stops. The next page measures estimator calibration offsets and controlled image changes.

The opt-in --verify-terminal path now renders the east camera after control stops. It reports the fresh image estimate separately; it never changes the controller stop or MuJoCo physical score. Missing markers make the estimate unavailable.

Calibration error changes the visual decision

100 held-out paired trials per condition; each verifier stress leaves the reaching controller unchanged.

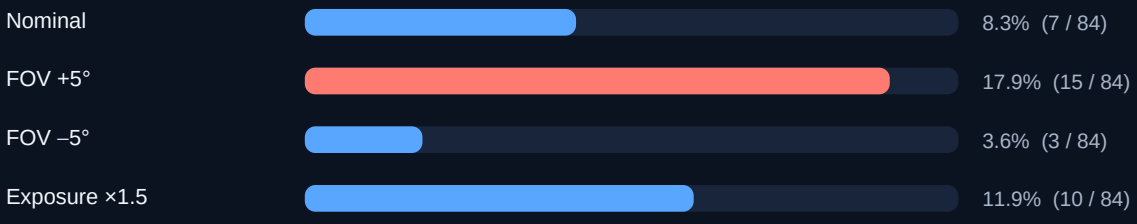
100
paired trials per condition

0 / 8
controller outcome changes

85 / 100
physical reaches in every row

Verifier condition	Both seen / estimate	False +	Missed	P95 error
Nominal	99 / 100	0	7	1.74 cm
Estimated FOV +5°	99 / 100	0	15	2.12 cm
Estimated FOV –5°	99 / 100	1	3	1.66 cm
Camera world X +2 cm	99 / 100	0	7	1.74 cm
Estimated yaw +3°	99 / 100	0	5	1.84 cm
Exposure ×0.25	0 / 100	n/a	n/a	n/a
Exposure ×1.5	99 / 100	0	10	2.52 cm
Target occluded	0 / 100	n/a	n/a	n/a
Tip occluded	0 / 100	n/a	n/a	n/a

MISSED PHYSICAL SUCCESSES AMONG AVAILABLE ESTIMATES



What these stresses do and do not establish

A +5° FOV estimate doubled missed successes (7 to 15); –5° caused one false success. Dim exposure and marker occlusion yielded no estimate, so those trials have no decision score. Brightness is a uniform post-render multiplier and occlusion uses marker-color removal plus inpainting. This is a simulator stress test, not real-camera evidence.

The quality gate trades coverage for fewer errors

A heuristic marker score was swept on the same held-out trials; these results do not select a deployment threshold.

01 **Contour area**

02 **Circularity**

03 **Saturation**

04 **Brightness**

Score = minimum component across both detected markers; it is a heuristic, not a probability.

Minimum score	Coverage	False +	Missed	Condition
0.00	99 / 100	1	3	FOV estimate –5°
0.15	93 / 100	1	3	FOV estimate –5°
0.20	39 / 100	1	3	FOV estimate –5°
0.50	9 / 100	0	1	FOV estimate –5°

AVAILABLE ESTIMATES AFTER APPLYING THE GATE



A useful abstention point has not been found

The 0.50 gate removed the single false success under a –5° FOV estimate, while leaving 9% coverage. At 0.20, coverage was 39% and the false success remained. Scores do not reveal calibration bias from marker appearance; no threshold is selected.

The frozen quality gate failed its coverage target

Threshold chosen on seeds 200-204, then evaluated unchanged on seeds 300-304; each split has 100 target trials paired across nine image conditions.

Calibration rule

Choose maximum candidate coverage with zero observed false-success decisions; require at least 50% coverage for a usable gate.

INDEPENDENT VALIDATION RESULTS

Gate	Accepted / candidate	Coverage	False +	Missed
Ungated 0.00	600 / 600	100%	10	48
Frozen 0.60	23 / 600	3.8%	0	3

CANDIDATE-ESTIMATE COVERAGE



The gate acted as a brightness filter

All 23 accepted estimates came from bright exposure. It accepted none from nominal images or any camera calibration shift. Across all 900 validation condition-trials, 23 estimates were available (2.6%); dim and occluded images had no candidate detection before gating.

Paired held-out video

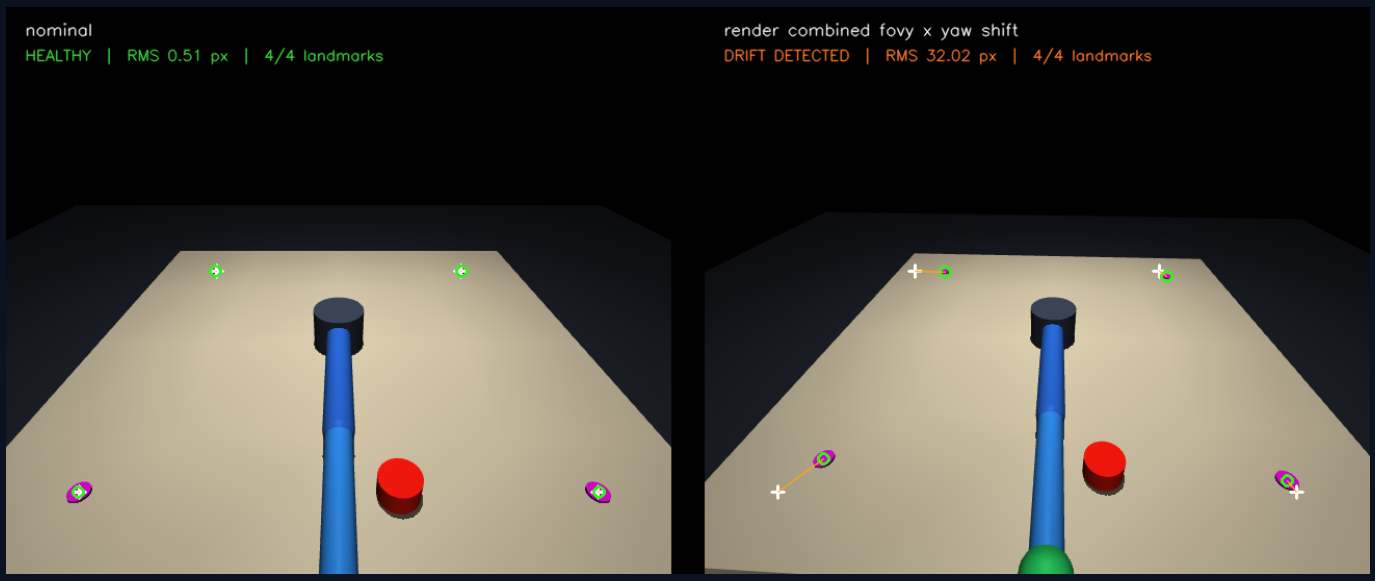
Seed 302, episode 17: fresh-view estimate 4.494 cm, MuJoCo physical error 4.6 cm. The 0.60 gate abstains; both clips preserve the same reaching outcome.

Negative result: no usable threshold found

The gate removed 10 false visual success decisions on validation but retained only 3.8% of candidate estimates, below the 50% usability floor. Zero observed errors do not establish a risk guarantee; no threshold is recommended.

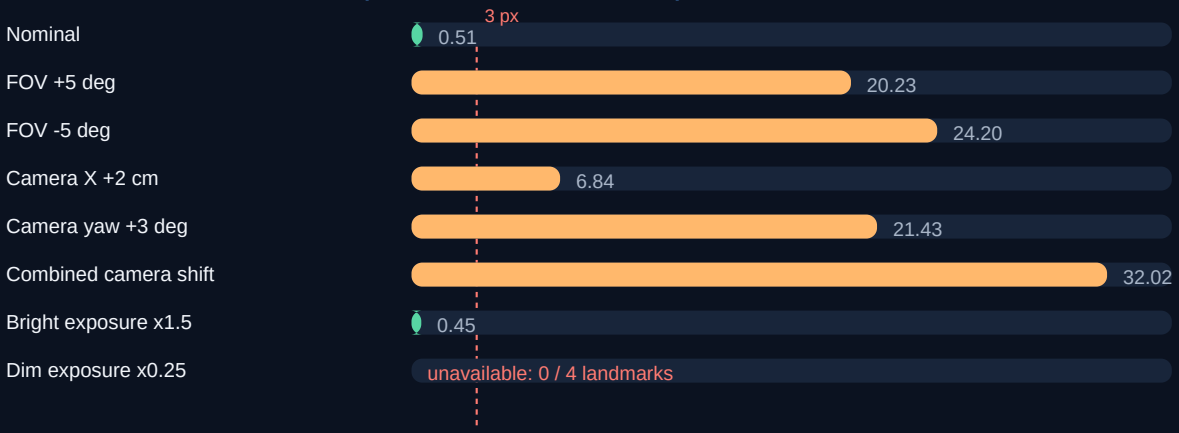
Independent landmarks expose camera-model drift

A separate simulator diagnostic compares known tabletop points with image detections while the rendered camera moves away from nominal calibration.



White crosses: nominal projection | Green rings: detected centers | Orange lines: residuals

MEASURED REPROJECTION RMS (PIXELS; ALARM LINE AT 3 PX)



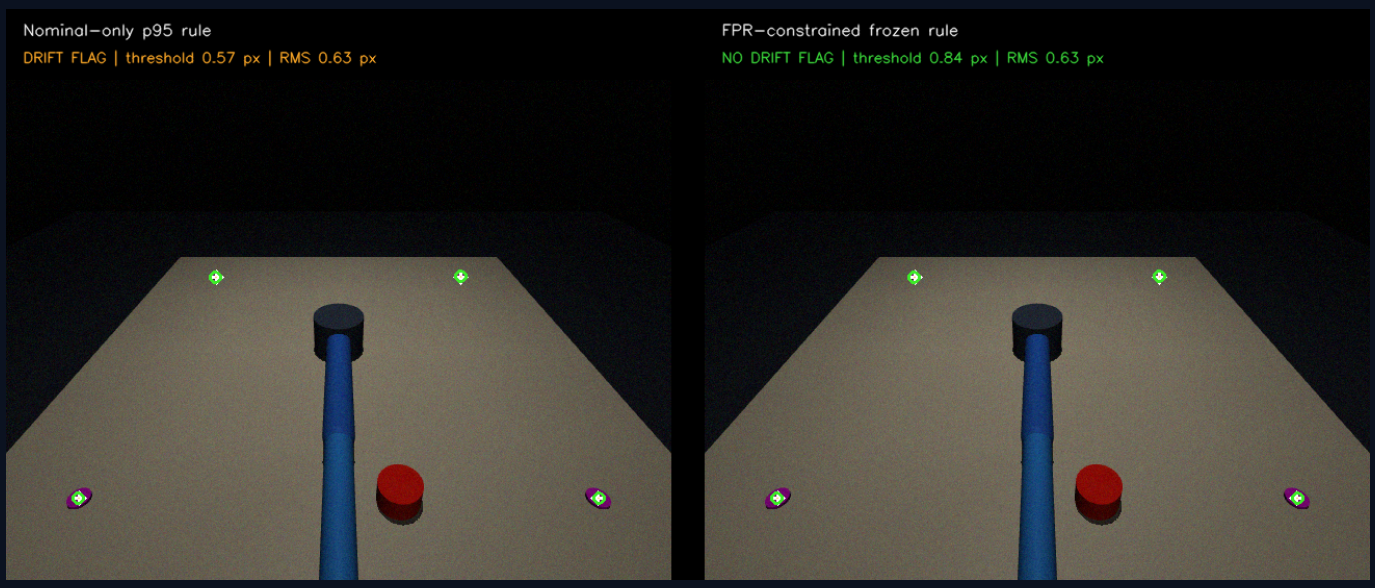
01 Known table coordinates → **02** Nominal pinhole projection → **03** Magenta center detections → **04** Match + pixel RMS residual

Promising detectability; threshold still exploratory

Five renderer-side camera shifts exceeded 3 px RMS. Nominal and bright-exposure images remained below it; dim images were unavailable. The 3 px threshold has not been validated on independent or real-camera frames. This diagnostic neither updates calibration nor changes reaching.

Labelled calibration shifts improved the threshold

A threshold was selected on seeds 400-404, frozen, then evaluated on fresh seeds 500-504 with broader image noise, brightness, and blur.



The same held-out nominal frame triggers the lower threshold but not the frozen supervised threshold.

HELD-OUT RESULTS (AVAILABLE LANDMARK ESTIMATES)

Threshold policy	RMS px	Nominal false alarms	Subtle	Moderate	Severe
Nominal-only p95	0.573	7 / 95 (7.4%)	100 / 100	100 / 100	100 / 100
Supervised, 5% cap	0.841	0 / 95	100 / 100	100 / 100	100 / 100
Fixed exploratory	3.000	0 / 95	60 / 100	93 / 100	100 / 100

CALIBRATION RULE

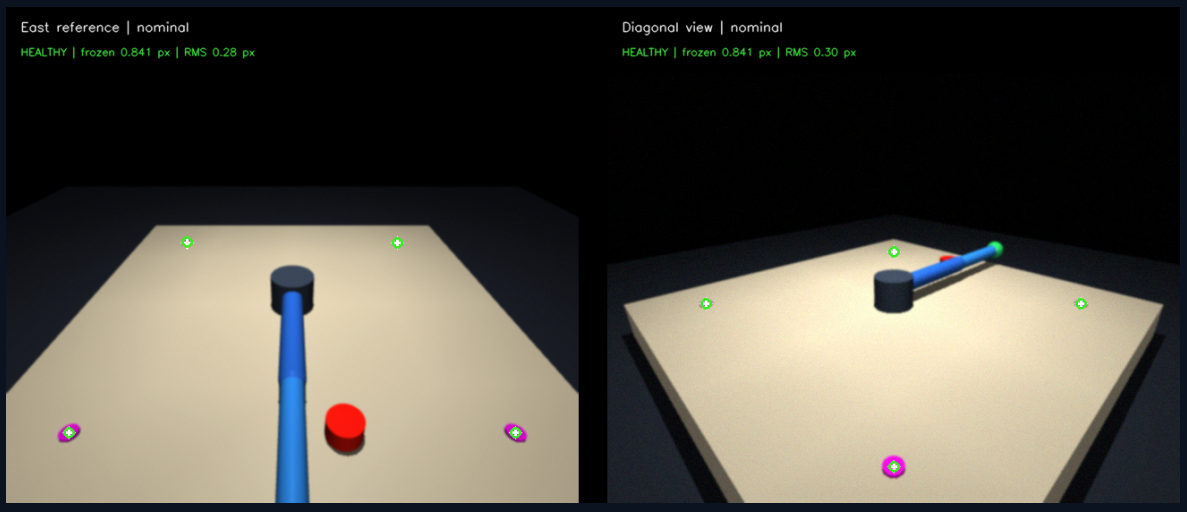
The supervised rule maximized shifted-camera detection on labelled calibration renders while keeping nominal calibration false alarms at or below 5%. It selected 0.841 px, with 0/100 nominal false alarms and 300/300 calibration shifts detected.

Promising result with a narrow validation boundary

On held-out seeds, the supervised rule flagged 0/95 nominal images with visible landmarks and detected all 300 new camera shifts. Five dim probes were unavailable. The study reuses one scene, landmark layout, shift families, and severity ranges; it is not a real-camera or reaching-success result.

Camera pose changes the residual evidence

The 0.841 px East-view threshold was frozen and tested on 3,000 simulator renders across five camera poses and synthetic radial lens warps.



White crosses: expected centers | Green rings: detections | Orange lines: candidate residuals

FROZEN-THRESHOLD CAMERA-SHIFT RESULTS

Pose	Nominal alarms	Subtle	Moderate	Severe / unavailable
East reference	0/95	100/100	100/100	100/100; 0 unavailable
West lower	2/95	100/100	99/99	82/82; 18 unavailable
North higher	0/95	99/100	100/100	100/100; 0 unavailable
Diagonal	0/95	100/100	100/100	99/99; 1 unavailable
East grazing	Unavailable	Unavailable	Unavailable	Unavailable

RADIAL DISTORTION ALARMS (25 IMAGES PER COEFFICIENT)

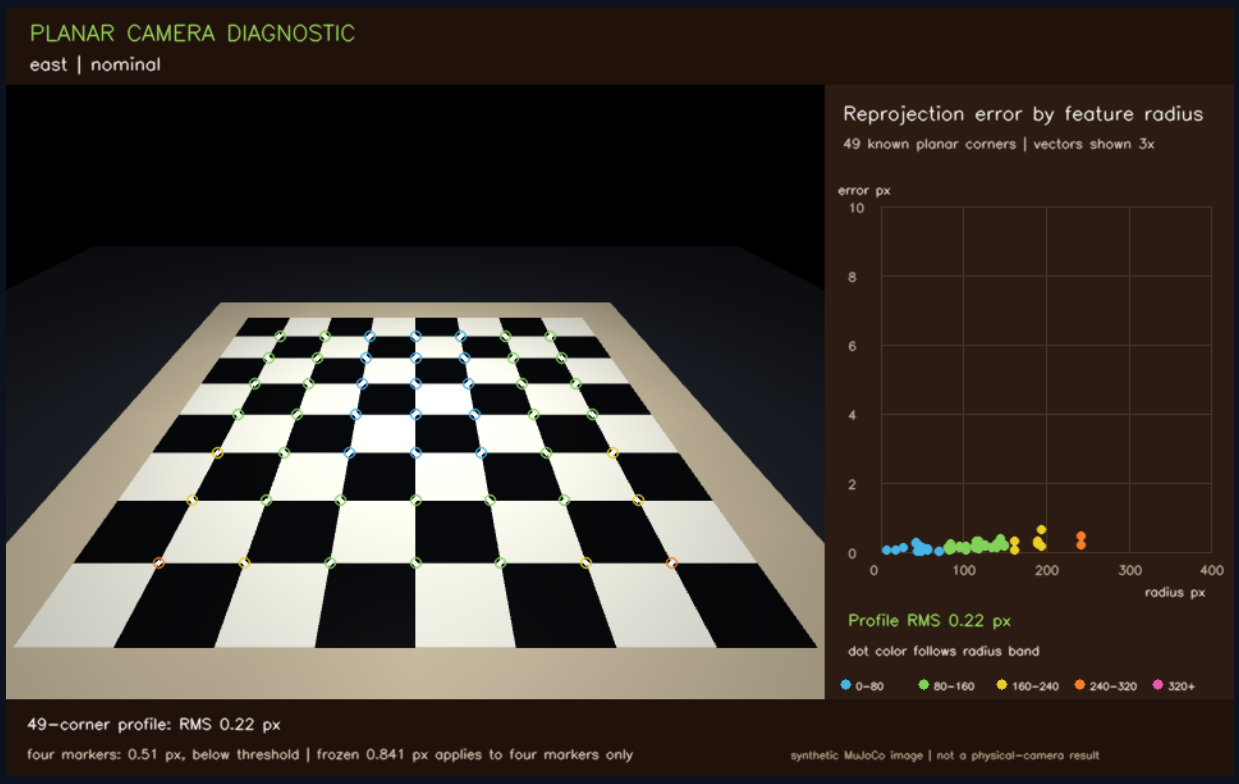
Synthetic k1	East	North higher	Southwest diagonal
–0.005	0/25	0/25	0/25
+0.005	25/25	0/25	0/25
+0.010	25/25	0/25	1/25
+0.020	25/25	25/25	25/25

A frozen pixel threshold does not transfer uniformly

The same +0.010 radial warp triggered 25/25 East images but 0/25 higher-North images. Edge-clipped expected points produce an unavailable result; their raw candidate RMS is retained. A dense planar follow-up now profiles residuals by radius. Next: validate it on recorded camera frames.

Residuals now have a radius

A 49-corner planar target adds spatial coverage to the frozen four-marker camera check. The profile is descriptive; the 0.841 px threshold remains specific to the original markers.



Color marks expected image radius; white dots show detected corners; residual vectors are exaggerated 3x.

NOMINAL SIMULATOR COVERAGE (FIVE APPEARANCE SAMPLES PER VIEW)

Pose	Grid available	Grid max radius	4-marker max radius	Grid median RMS
East	5/5	241 px	297 px	0.28 px
West lower	5/5	272 px	343 px	0.79 px
North higher	5/5	186 px	223 px	0.16 px
Southwest diagonal	4/5	182 px	210 px	0.41 px
East grazing	3/5	287 px	366 px	1.59 px

Radius explains a view-dependent miss

East k1=+0.010: median signed radial residual is -0.05 px at 0-80 px and +0.86 px at 240-320 px. The frozen four-marker rule alarms on 5/5 East images, then 0/5 higher-North images at the same synthetic coefficient.

More field coverage costs visibility

A 0.17 m-square probe reached a 354 px radius but fell to 2/5 West, 4/5 diagonal, and 0/5 grazing availability. The 0.15 m-square target keeps higher availability while undersampling some edge radii. All evidence remains synthetic.

Take the diagnostic to recorded images

The simulator experiment maps residuals by feature radius and separates target coverage from the legacy four-marker decision. Physical camera evidence is the next step.

Completed milestone: 325 paired diagnostic cases

Five camera poses, 13 nominal/drift/lens conditions, and five held-out appearance seeds. The dense profile stays descriptive; the frozen 0.841 px threshold is applied only to the original four markers. The wider target probe is preserved as a negative result.

Next: capture and validate independent camera evidence

- Record the checkerboard, camera intrinsics, and target pose in a manifest.
- Capture nominal and shifted frames at two or more camera poses.
- Move the target through center and edge radii; report clipping separately.
- Keep capture, calibration, and validation images in disjoint groups.
- Keep physical reaching separate from fresh visual confirmation.
- Keep the terminal verifier and quality gate opt-in until evidence supports them.

REPRODUCE THE MAIN REPORTS

```
uv run pytest -q
uv run visual-servo-camera-calibration-health --output artifacts/camera-calibration-health.json
uv run visual-servo-camera-calibration-validation --output artifacts/camera-calibration-health-validation.json
uv run visual-servo-camera-calibration-transfer --output artifacts/camera-calibration-transfer-validation.json
uv run visual-servo-camera-calibration-planar --output artifacts/camera-calibration-planar-profile.json
uv run python scripts/generate_camera_calibration_planar_demo.py
uv run python scripts/generate_camera_calibration_planar_design_demo.py
```

Video and source map

assets/demos/ contains control, calibration, latency, prediction, and verification examples. camera_calibration_planar.mp4 shows radius error and the separate frozen four-marker decision; camera_calibration_planar_target_design.mp4 replays the larger-grid availability failure. Start with camera_calibration_planar.py, camera_calibration_transfer.py, camera_calibration_health.py, terminal_verification.py, controller.py, and run.py.

Compare the figures with README.md and the JSON reports under artifacts/; those reports are local benchmark evidence and can be regenerated. The calibration diagnostic measures image-to-model disagreement, while the reaching score remains a separate MuJoCo measurement. Current camera-calibration evidence is synthetic; recorded camera frames remain necessary for physical claims.

Publication boundary

This work is local and unpushed. Review the complete diff and evidence before any publication.